Graph Theory

What is a graph?

A graph G is a tuple, i.e, $G := (V, E)$, where V (or $V(G)$) is a set of elements called the vertex set (or node) (each element in this set is called a vertex (or node)) and E (or $E(G)$) is a collection of 2-sized subsets of V and is called the edge set (each element in E is called an edge).

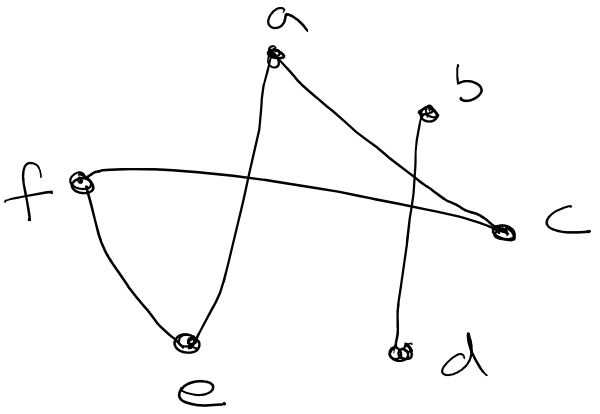
Example

$G = (V, E)$, where

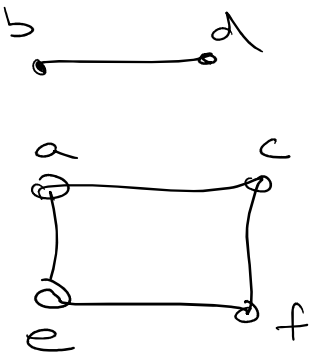
$$V = \{a, b, c, d, e, f\}$$

$$E = \{\{a, c\}, \{a, e\}, \{b, d\}, \{e, f\}, \{c, f\}\}$$





OR



A more general notion

Hypergraph → is a tuple

$H = (V, E)$

V → vertex set (a collection of elements ~~2-sized~~)

E → a collection of subsets of V.
(a subset in E is called a hyperedge)

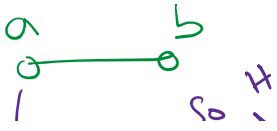
Example $H = (V, E)$

$V = \{a, b, c, d, e, f\}$

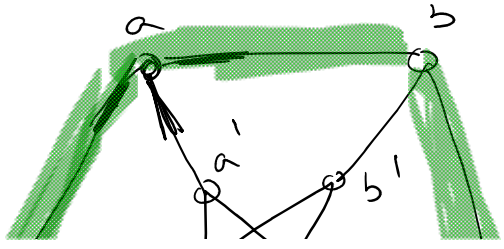
$E = \{\{a, b, c, d\}, \{c, e, f\}, \{a, e, f\}, \{b, d\}\}$

Some definitions on graphs

$H = (V', E')$

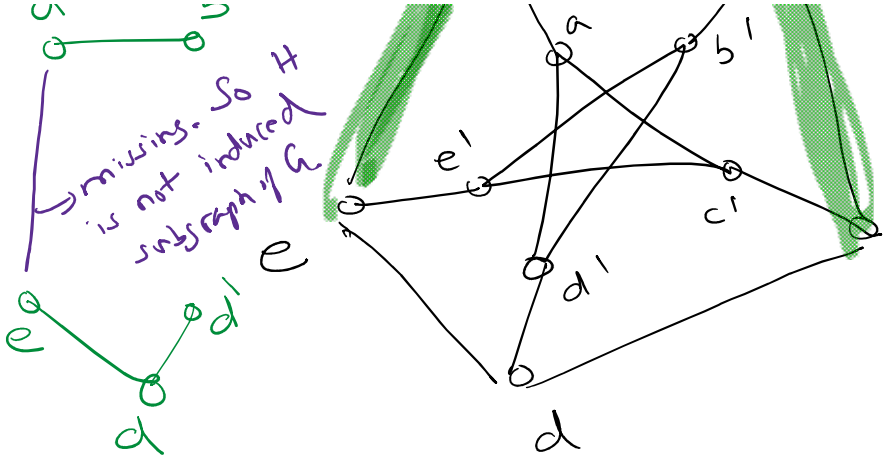


So H



$G = (V, E)$

$V = \{a, b, c, d, e, a', b', c'\}$



$V = \{a, b, c, d, e, a', b', c', d', e'\}$

$E = \{\{a,b\}, \{b,c\}, \{c,d\}, \{d,e\}, \{e,a\}$
OR
(a,b) $\{a,a'\}, \{b,b'\}$
OR
ab $\{e,e'\}$
$\{a',c'\}, \{c',e'\}$
$\{e',b'\}, \{b',d'\}$
$\{d',a'\}$ $\}$

$|V| = 10$

$|E| = 15$

a is a neighbour of b

$N_G(a) = \{b, a', e\}$

Notion

(i) degree of a vertex $\longrightarrow$ no. of edges it is present in.

$$\deg_G(a) = 3, \quad \deg_G(b) = 3$$

(ii) maximum degree of G, denoted by $\Delta(G)$, is

$$\Delta(G) := \max\{\deg_G(v) : v \in V(G)\}$$

above example, $\Delta(G) = 3$.

(iii) minimum degree of G, denoted by

$$\delta(G) := \min\{\deg_G(v) : v \in V(G)\}$$

$\delta(G) := \min\{ \deg_G(v) : v \in V(G) \}$

above example, $\delta(G) = 3$

(iv) given two vertices u and v, we say u is adjacent to v if $\{u, v\} \in E(G)$.

u is a neighbor of v in G if $\{u, v\} \in E(G)$.

(v) Neighborhood of a vertex v in G is denoted $N_G(v)$ and is defined as $N_G(v) = \{ u : \{u, v\} \in E(G) \}$

(vi) $H = (V', E')$ is a subgraph of $G = (V, E)$ if $V' \subseteq V$ and $E' \subseteq E$.

(vii) a vertex v is incident with an edge e if $v \in e$.

edge e if $v \in e$.

↳ 2-sized subset of V.

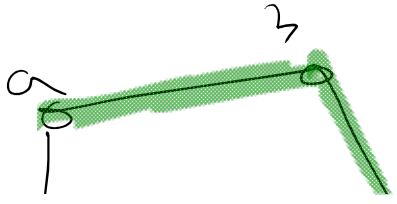
(ix) $H = (V', E')$ is an induced subgraph of $G = (V, E)$ if $V' \subseteq V$ and $E' = \{(u,v) \in E(G) : u, v \in V'\}$

(x) $H = (V', E')$ is a spanning subgraph of $G = (V, E)$ if H is a subgraph of G and $V' = V$.

↳ spanning

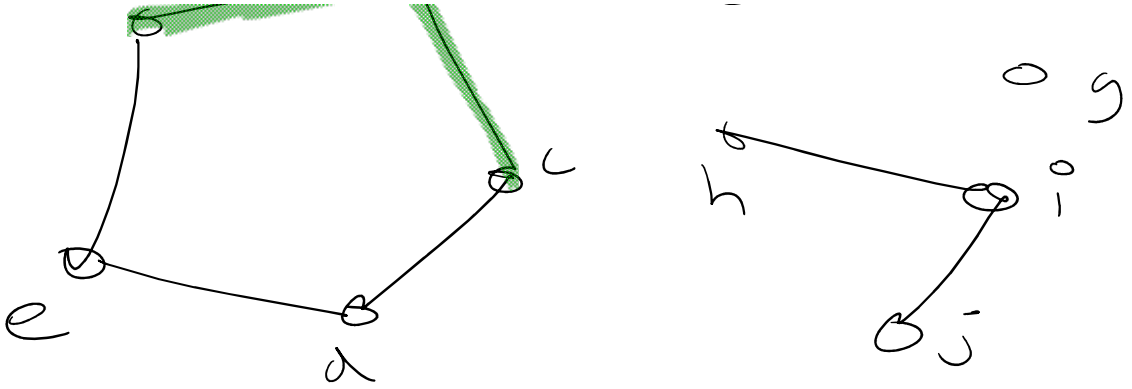
(xi) Average degree of a graph G, denoted $d_{avg}(G)$ is

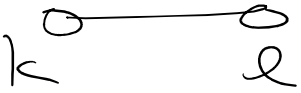
$$d_{avg}(G) := \frac{\sum_{v \in V(G)} deg_G(v)}{|V|}$$



for $H = (V, E)$

a



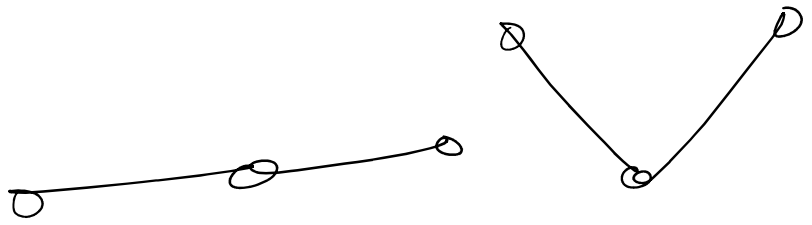


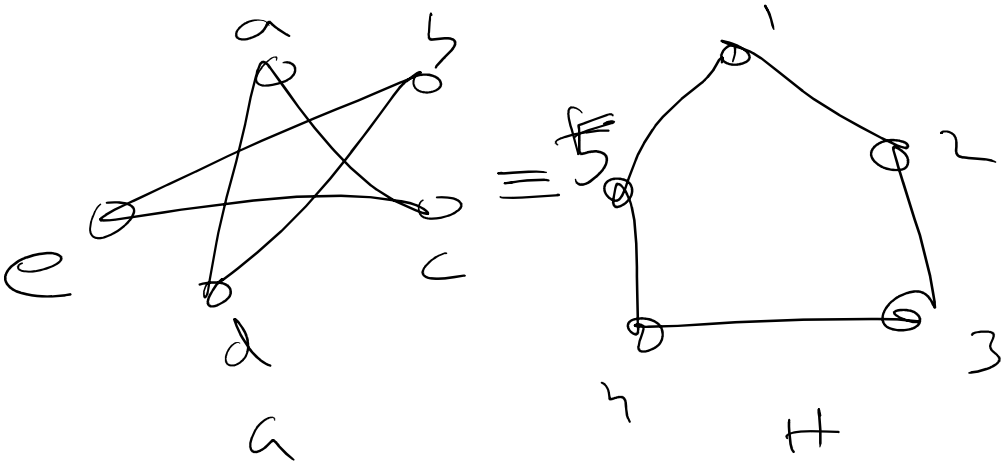
$V(H) = \{a, b, \ldots, l\}$

$E(H) = \{\{a,e\}, \{a,b\} \ldots \ldots \{k,l\}\}$

* sum of degrees $= 2 \cdot |E(G)|$.

Therefore no. of odd degree vertices is even.





G is isomorphic to H.

$f : V(G) \to V(G)$ as

$f(a)=1$, $f(c)=2$, $f(e)=3$, $f(b)=4$,

$f(d)=5$

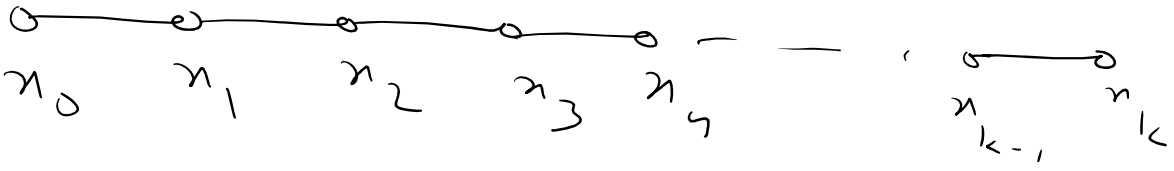
* A <u>path</u> is a non-empty graph

$P=(V, E)$, where

$V=\{x_0, x_1, x_2, \ldots, x_k\}$

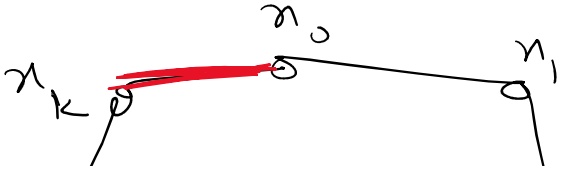
$E=\{\{x_0,x_1\}, \{x_1,x_2\}, \{x_2,x_3\}, \ldots$

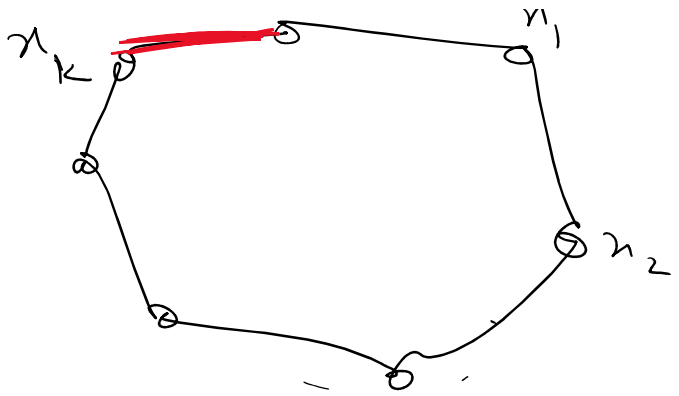
$\ldots \{x_{k-1}, x_k\}\}$



<u>length</u> of a path is the no. of edges in it.

* Cycle





Let $P = (V, E)$ be a path,

where $V = \{x_0, x_1, \ldots, x_k\}$

$E = \{\{x_0, x_1\}, \{x_1, x_2\}, \ldots, \{x_{k-1}, x_k\}\}$

Then, $P + \{x_k, x_0\}$ is a cycle